A Non-Invasive Smart Error Alert System System for Legacy Industrial Machines Machines

Paper Id:692

THE PROBLEM

Legacy machines can't communicate faults remotely

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Manual inspection only

Fault info shown on local LEDs / LCDs — someone must be present

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Delayed fault detection

Missed faults cause longer downtime and higher maintenance costs

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Costly upgrade options

PLC retrofitting & sensor installs need expensive hardware changes

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Brownfield challenge

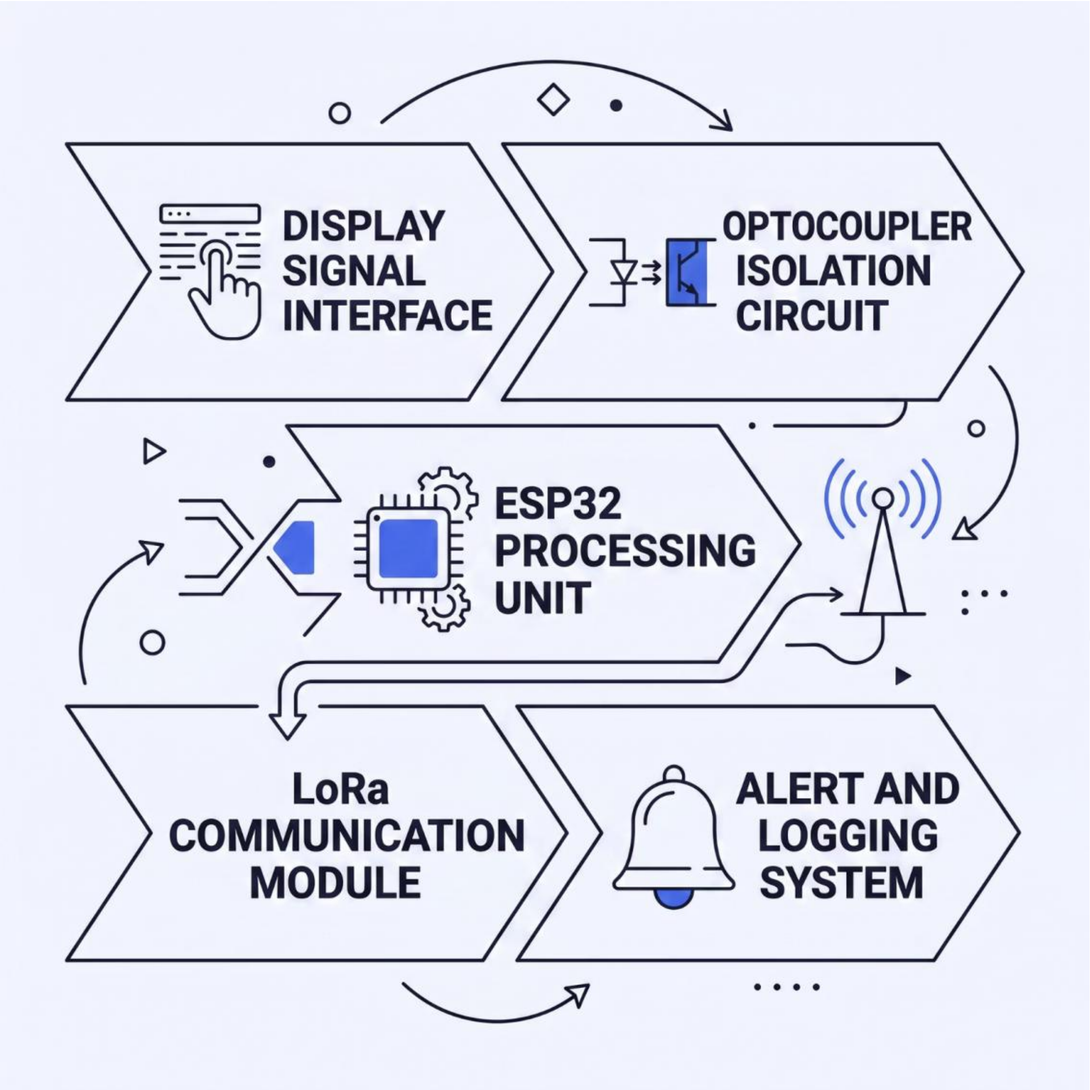
Machines pre-date IIoT — no digital outputs or remote diagnostics

OBJECTIVE:
Enable real-time fault alerts

The main objective is to develop a low-cost, non-invasive, and electrically safe monitoring system that can detect machine faults remotely without changing the original machine hardware.

- ✓ No PLC changes
- ✓ No extra sensors
- ✓ No cameras
- ✓ No machine downtime

SOLUTION



Galvanic isolation

Optocoupler protects embedded system from machine voltage spikes

Pattern lookup table

Binary display states mapped to specific fault codes

LoRa Communication

Long-range, low-power radio transmission delivers fault alerts across large factory floors reliably

HARDWARE & SOFTWARE

Hardware

Microcontroller	ESP32 (dual-core, GPIO, low-power)
Isolation	Optocoupler (PC817)
Communication	LoRa module , GSM
Power supply	Regulated 5V / 3.3V subsystem

Software

Language	Embedded C
IDE	Arduino IDE
Wireless protocol	LoRa
Fault method	Pattern lookup table
Interfaces	SPI, UART

EXPERIMENTAL RESULTS

97.2%

Fault detection accuracy

vs 85–88% existing

1.1 s

Average alert latency

vs 4–6 s existing

2.4%

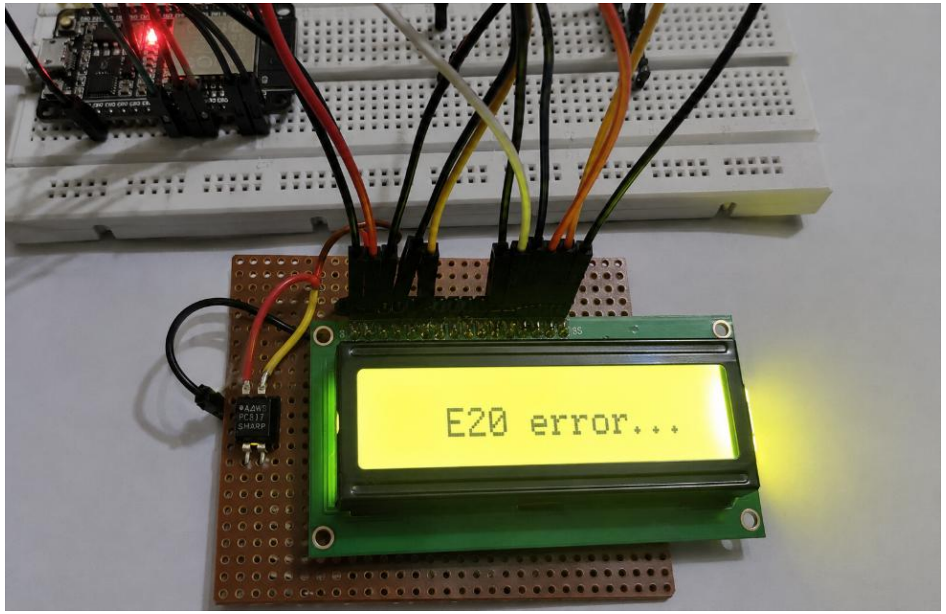
False alert rate

vs 8–12% existing

98.1%

Wireless transmission success

vs 85–90% existing



Proposed system vs existing methods

Method	HW Modification	Cost	Industrial Reliability	Legacy Fit
PLC Retrofit	High	High	High	Moderate
Sensor-Based	Moderate	Moderate	Moderate	Moderate
Camera / OCR	Low	Moderate	Low–Moderate	Moderate
Cloud IIoT	Moderate	High	Moderate	Moderate
Proposed (Optocoupler)	Very Low	Low	High	High

Advantages

- Non-invasive monitoring
- Low implementation cost
- Real-time fault alerts
- Easy integration with legacy machines

Future Scope

In future, this system can be integrated with cloud dashboards, mobile applications, centralized monitoring platforms, and machine learning-based fault prediction systems.

Limitations

- Requires accessible display signal lines
- Machine-by-machine calibration needed
- Multiplexed displays need advanced decoding
- Focused on fault states, not full analytics

CONCLUSION

- **Non-invasive** — no changes to original machine hardware
- **Electrically safe** — optocoupler galvanic isolation
- **Real-time** — 1.1s average alert latency
- **Scalable** — modular firmware, easy machine-specific mapping
- **Economical** — fraction of the cost of PLC retrofit or replacement
- **Thank you — Questions welcome**